Abdulla **Ansari**

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Summary

Physics undergraduate with strong C++ and Python programming ability and hands-on experience in computational physics and 3D simulation development. Skilled in building and optimising C++ simulation systems for particle propagation and radiation transport modelling using GEANT4 within Linux-based workflows, grounded in a solid foundation across classical mechanics, nuclear physics, and advanced mathematical methods. Proven ability to deliver technically precise, well-documented solutions independently across demanding technical projects.

Experience

Freelance Programming Developer — *Remote*
Independent — *Jan. 2021 - Present*

- Built data-driven tools and automated systems in C++ and Python for client projects across diverse technical environments.
- Designed cross-platform components and backend logic with structured, precise problem-solving, delivering robust, well-documented solutions independently.

Private Tutor — *Remote*
Independent — *Sept. 2020 - Present*

- Delivered one-to-one instruction in Physics, Mathematics, and Computer Science, explaining technical concepts clearly to varied audiences.
- Developed tailored, step-by-step solution strategies while managing all scheduling and client communication independently.

Relationship Manager — *Dubai, UAE*
Bellavista Real Estate — *Feb. 2023 - Jul. 2023*

- Managed property listings and communication between buyers, sellers, agents, and international agencies, developing strong negotiation and communication skills in a fast-paced environment.

Projects

Optimization Of X-Ray Imaging Using GEANT4 For Enhanced Field-Of-View — *Debrecen, Hungary*
Bachelor Thesis — *Dec. 2025 - Present*

- Designed a 3D multi-source simulation environment in GEANT4 using C++ and Python, modelling radiation transport and particle interactions.
- Built C++ and Python simulation logic for source geometry, photon propagation, and material interactions, optimising performance.
- Applied statistical analysis to benchmark and refine geometry, managing the full lifecycle with Git/GitHub.

F1 Calculator & Data Tracker
Personal Project — *Apr. 2026*

- Built an F1 calculator and data tracker in Python using FastAPI to serve driver performance data through a lightweight web API.
- Used Matplotlib to visualise and plot driver statistics, presenting race and performance data in clear, interpretable charts.

Web Messaging Chat Room
Personal Project — *Nov. 2025*

- Built a real-time web chat room in Python using sockets and threading to let multiple clients connect and exchange messages.
- Developed a simple HTML front-end and a multithreaded server to handle concurrent client connections and simultaneous users.

Offshore Wind Turbine Simulation — *Dubai, UAE*
Personal Project — *May 2022 - Jul. 2022*

- Designed and implemented a Monte Carlo simulation in Python, applying probabilistic methods to model complex multi-parameter system behaviour under variable conditions.
- Built multi-parameter computational models incorporating physical and environmental variables, validating outputs against theoretical benchmarks through iterative refinement.

Education

Bachelor of Science - B.Sc Physics — *Debrecen, Hungary*
University of Debrecen — *Sept. 2023 - Jan. 2027*

- Strong foundation in classical mechanics, electrodynamics, quantum and nuclear physics, and advanced mathematical methods for physics.
- Computational physics and simulation development in C, Python, GEANT4, and Linux, with coursework in C/Python Programming, Computational Physics, and Fundamental Interactions.

Skills

Programming	C++, Python, C, SQL, HTML
Simulation Tools & Methods	GEANT4, Houdini, SciLab, Linux, Git/GitHub, Data analysis, Statistical modeling
Languages	English (C2), Hindi (B1), Italian (A2), German (A1)